

Data Structure and algorithms (LAB)

Lab Task

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PROBLEM 1

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    string productID;
```

```
    Node *next;
```

```
    Node(const string &id) : productID(id), next(nullptr) {}
```

```
};
```

```
Node *head = nullptr;
```

```
Node *current = nullptr;
```

```
int main()
```

```
{
```

```
    Node *firstNode = new Node("p101");
```

```
    Node *secondNode = new Node("p102");
```

```
    Node *thirdNode = new Node("p103");
```

```
    Node *fourthNode = new Node("p104");
```

```
    firstNode->next = secondNode;
```

```
    secondNode->next = thirdNode;
```

```
thirdNode->next = fourthNode;
```

```
fourthNode->next = nullptr;
```

```
head = firstNode;
```

```
current = head;
```

```
while (current->next != nullptr)
```

```
{
```

```
    cout << current->productID << " -> ";
```

```
    current = current->next;
```

```
}
```

```
cout << current->productID << " -> ";
```

```
cout << "nullptr" << endl;
```

```
Node *fifthNode = new Node("p105");
```

```
fifthNode->next = nullptr;
```

```
current->next = fifthNode;
```

```
current = head;
```

```
while (current->next != nullptr)
```

```
{
```

```
    cout << current->productID << " -> ";
```

```
    current = current->next;
```

```
}
```

```
cout << current->productID << " -> ";
```

```
cout << "nullptr" << endl;
```

```
}
```

```
Node *firstNode = new Node("Lahore");
```

```

Node *secondNode = new Node("Gujranwala");
Node *thirdNode = new Node("Rawalpindi");
Node *fourthNode = new Node("Islamabad ");

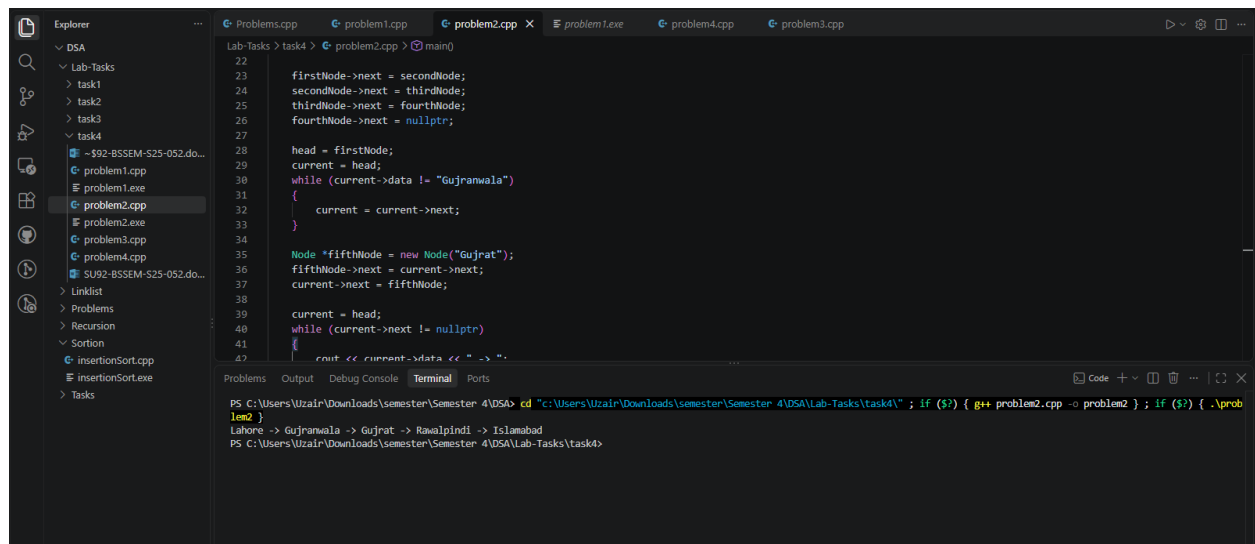
firstNode->next = secondNode;
secondNode->next = thirdNode;
thirdNode->next = fourthNode;
fourthNode->next = nullptr;

head = firstNode;
current = head;
while (current->data != "Gujranwala")
{
    current = current->next;
}

Node *fifthNode = new Node("Gujrat");
fifthNode->next = current->next;
current->next = fifthNode;

current = head;
while (current->next != nullptr)
{
    cout << current->data << " -> ";
    current = current->next;
}
cout << current->data;
}

```



PROBLEM 3

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
struct Node
```

```
{
```

```
    string productID;
```

```
    Node *next;
```

```
    Node(const string &id) : productID(id), next(nullptr) {}
```

```
};
```

```
Node *head = nullptr;
```

```
Node *current = nullptr;
```

```
int main()
```

```
{
```

```
    Node *firstNode = new Node("O101");
```

```
Node *secondNode = new Node("O102");  
Node *thirdNode = new Node("O103");  
Node *fourthNode = new Node("O104");
```

```
firstNode->next = secondNode;  
secondNode->next = thirdNode;  
thirdNode->next = fourthNode;  
fourthNode->next = nullptr;
```

```
head = firstNode;  
current = head;  
while (current->next != nullptr)  
{  
    cout << current->productID << " -> ";  
    current = current->next;  
}  
cout << current->productID << " -> ";  
cout << "nullptr" << endl;
```

```
// Inserting new Node at the end  
Node *fifthNode = new Node("O105");  
fifthNode->next = nullptr;  
current->next = fifthNode;
```

```
// Inserting new Node after O102  
current = head;  
while (current->productID != "O102")  
{  
    current = current->next;
```

```

}

Node *sixthNode = new Node("O1025");

sixthNode->next = current->next;

current->next = sixthNode;

current = head;

while (current->next != nullptr)
{
    cout << current->productID << " -> ";

    current = current->next;
}

cout << current->productID << " -> ";

cout << "nullptr" << endl;

}

```

The screenshot shows a Visual Studio Code editor with a C++ project. The file explorer on the left shows a directory structure with files like `problem1.cpp`, `problem2.cpp`, `problem3.cpp`, and `problem4.cpp`. The main editor window displays the code for `problem3.cpp`, which includes a linked list implementation. The code defines a `Node` structure and a function to insert a new node at the end of the list. The terminal output shows the execution of the program, displaying the product IDs of the nodes in the list.

```

23 firstNode->next = secondNode;
24 secondNode->next = thirdNode;
25 thirdNode->next = fourthNode;
26 fourthNode->next = nullptr;
27
28 head = firstNode;
29 current = head;
30 while (current->next != nullptr)
31 {
32     cout << current->productID << " -> ";
33     current = current->next;
34 }
35 cout << current->productID << " -> ";
36 cout << "nullptr" << endl;
37
38 // Inserting new Node at the end
39 Node *fifthNode = new Node("O105");
40 fifthNode->next = nullptr;
41 current->next = fifthNode;
42

```

Terminal Output:

```

PS C:\Users\Uzair\Downloads\Semester\Semester 4\DSA\Lab-Tasks\task4> cd "c:\Users\Uzair\Downloads\Semester\Semester 4\DSA\Lab-Tasks\task4" & if ($?) { g++ problem3.cpp -o problem3 } ; if ($?) { .\problem3 }
0101 -> 0102 -> 0103 -> 0104 -> nullptr
0101 -> 0102 -> 01025 -> 0103 -> 0104 -> 0105 -> nullptr
PS C:\Users\Uzair\Downloads\Semester\Semester 4\DSA\Lab-Tasks\task4>

```

PROBLEM 4

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
struct Node
{
    string productID;
    Node *next;

    Node(const string &id) : productID(id), next(nullptr) {}
};
```

```
Node *head = nullptr;
Node *current = nullptr;
```

```
int main()
{
    Node *firstNode = new Node("E101");
    Node *secondNode = new Node("E102");
    Node *thirdNode = new Node("E103");
    Node *fourthNode = new Node("E104");

    firstNode->next = secondNode;
    secondNode->next = thirdNode;
    thirdNode->next = fourthNode;
    fourthNode->next = nullptr;

    // Inserting new Node after E102
    head = firstNode;
    current = head;
    while (current->productID != "E102")
    {
        current = current->next;
```

```
}
```

```
Node *fifthNode = new Node("E1025");
```

```
fifthNode->next = current->next;
```

```
current->next = fifthNode;
```

```
// Inserting new Node after E103
```

```
current = head;
```

```
while (current->productID != "E103")
```

```
{
```

```
    current = current->next;
```

```
}
```

```
Node *sixthNode = new Node("E1035");
```

```
sixthNode->next = current->next;
```

```
current->next = sixthNode;
```

```
current = head;
```

```
while (current->next != nullptr)
```

```
{
```

```
    cout << current->productID << " -> ";
```

```
    current = current->next;
```

```
}
```

```
cout << current->productID << " -> ";
```

```
cout << "nullptr" << endl;
```

```
}
```

